COMPUTER FORENSICS CASE STUDY REPORT

SBT-DF204 — Case Study 5

Investigating DJI Mavic Air Flight Evidence

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Registration Number: 2025/FWSD/11378

Course: SBT-DF204 Computer Forensics Case Studies

Case Reference: SBT-DF204-C5

Evidence source: Android logical extraction of DJI GO 4 (dji.go.v4) application data

Analysis Workstation: Kali Linux VM

Report Date: 2 November 2026

1. Case Reference and Student Identification

Field	Value
Full name	Fasinu Temidayo Lucky
Registration number	2025/FWSD/11378
Course code	SBT-DF204 — Computer Forensics Case Studies
Assessment	Case Study 5 — Individual mobile and microSD forensic investigation
Case reference	SBT-DF204-C5
Case title	Investigating DJI Mavic Air Flight Evidence
Evidence examined in this report	Android_Logical.zip — logical extraction containing the DJI GO 4 (dji.go.v4) application data
Evidence referenced but not supplied	Raw internal microSD image (df048_internal_microSD.001) and EWF image (DF048.E01) — see Section 7
Analysis workstation	Kali Linux VM

This report is submitted individually. All interpretation, screenshots and evidence-log entries that follow were produced from my own direct examination of the supplied Android_Logical.zip extraction. This is an offline examination of authorised course artefacts; no drone, mobile device, account, cloud service, or removable media was flown, powered on, connected, synchronised, reset, or otherwise altered in the course of this examination.

2. Executive Summary and Conclusion

This case study examines an Android logical extraction (Android_Logical.zip, SHA-256 9a352b28...29285d, ~396 MB) containing the data folder of the DJI GO 4 application (dji.go.v4). Within that folder, a main flight record (TXT), two detailed flight records (DAT), a cached video (MP4) with its telemetry sidecar (.info), and six cached picture thumbnails were located, preserved and examined. The flight record, video filename, and .info sidecar together identify a DJI Mavic Air aircraft, serial-formatted string K1DF313BD4PR1, flown on 19 June 2018, with a second serial-formatted string (K4AEBQA3400DT) also present in the record. GPS coordinates recorded in the video's telemetry sidecar (39.9612, -106.2165) place the flight in the Rocky Mountain region of north-central Colorado, USA.

A consistent, internally-ordered sequence of filesystem timestamps (the TXT record, the cached MP4, and both DAT records being closed in that order, each a few seconds to a few minutes apart) supports a single flight-and-landing event on 19 June 2018, followed by at least one further device power cycle. However, a genuine, well-evidenced clock discrepancy of almost exactly two hours was found between the application's own self-recorded internal timestamp and the Android filesystem's timestamp for the same file (Section 5.3, Issue I-03) — this report builds its UTC timeline from the filesystem timestamps, which are internally self-consistent, while flagging the application-clock

family as uncorroborated for an absolute UTC anchor. The six cached picture thumbnails were independently found to carry filename-embedded dates in April 2018 — two months before this flight — demonstrating that the DJI GO 4 picture cache on this device is not evidence of the 19 June flight and must not be used to infer its content (Section 5.4, Issue I-02).

The raw internal microSD image and the EWF (Expert Witness Format) image referenced by the case brief — which would corroborate the mobile findings against the aircraft's own onboard storage — were not included in the material supplied for this examination (Section 7). No raw microSD or EWF findings are reported here, and this report does not cite figures (hashes, partition details, serial numbers) from secondary course material that could not be independently verified against actual image data in this examination. On the evidence that was available, this report can establish that the DJI GO 4 application on this Android device recorded data consistent with a single approximately four-minute flight on 19 June 2018 in Colorado, USA, followed by a further power cycle; it cannot establish who was operating the aircraft, and the cached video and pictures should not be assumed to represent the complete set of media captured during that flight. Section 6 sets out the full confidence assessment, and Section 8 documents every limitation and issue encountered, several of which (Issues I-02, I-03, I-06 and I-07) were found independently through direct inspection of the raw evidence rather than from any supplied summary.

3. Evidence Acquisition and Integrity (Investigation Question 1)

Task requirement: for each mobile and microSD source, record the source, original file name, byte size, SHA-256 value, working-copy method, tool version, and read-only or access precautions.

3.1 Source and working copy

Field	Value
Original file name	Android_Logical.zip
Size	396 MB (ls -lh; exact byte count was not separately captured)
SHA-256 (evidence copy)	9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d
SHA-256 (working copy, after cp)	9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d — identical
Working directory structure	~/SBT-DF204-Case5/{evidence,working,screenshots,notes} (mkdir -p)

Field	Value
Extraction path examined	working/Android_Extracted/Android_Logical/sdcard/DJI/dji.go.v4
Hash tool	sha256sum (GNU coreutils)

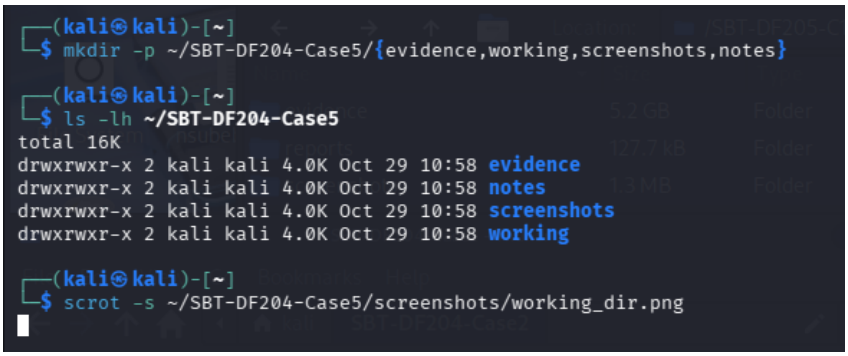


Figure 3.1 — Working directory structure created before any evidence was accessed, and the screenshot capture command used throughout this report.



Figure 3.2 — Android_Logical.zip in the evidence folder, 396 MB.

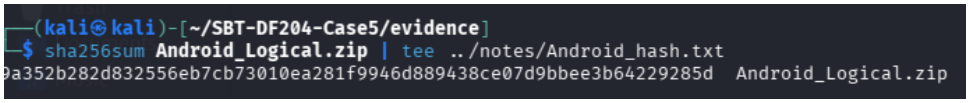


Figure 3.3 — SHA-256 of Android_Logical.zip computed in the evidence folder and recorded to notes/Android_hash.txt.

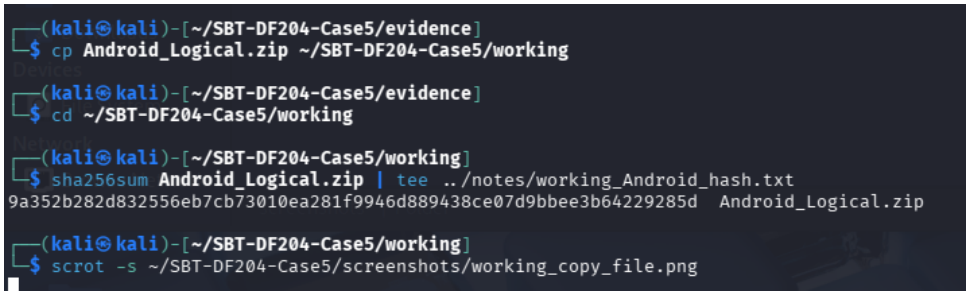


Figure 3.4 — Android_Logical.zip copied to the working folder and re-hashed; the SHA-256 value is identical to Figure 3.3, confirming a binary-identical working copy.

The original Android_Logical.zip was hashed in the evidence/ folder before any copy was made, then re-hashed after being copied to working/ — both values match exactly (9a352b28...29285d), confirming the working copy used for all further analysis is binary-identical to the preserved original. No individual SHA-256 values were separately captured for the TXT, DAT or .info files extracted from this archive; their integrity rests on the verified archive hash and on their internally consistent filesystem timestamps (Section 3.3, Issue I-05).

3.2 Extraction and folder structure

The zip was extracted to working/Android_Extracted/Android_Logical, and the DJI GO 4 application data folder was located at sdcard/DJI/dji.go.v4.



Figure 3.5 — The dji.go.v4 application data folder, listing 14 subdirectories including CACHE_IMAGE, DJI_RECORD and FlightRecord, the three folders examined in Section 5.

The dji.go.v4 folder contains (among others not examined further, as they fall outside the artefact families named in the brief): CACHE_IMAGE (cached picture thumbnails), databases, DJI_RECORD (cached video and its telemetry sidecar), DJI_SPALSH, editor, FlightRecord (main TXT and detailed DAT flight records), LOG, Package, RECORD_VOICE, VideoEditor, videoguide, visual_guide and xtScreenshots.

3.3 Evidence register and handling-log issues

A 10-entry Evidence_Register.csv and a Handling_Log.csv accompany this report. One handling issue is documented here because it bears directly on what this report can and cannot claim for Investigation Questions 7 and 8.

Issue I-01 — the case brief and a student working-note direct this report to cite two PowerPoint decks and a raw/EWF image that were not supplied with this examination's evidence package.

The case brief states that the raw microSD image (df048_internal_microSD.001) and EWF image (DF048.E01) should be examined directly, falling back to two named course slide decks (01_DJI_Mavic_Air_microSD_raw.pptx, 02_DJI_Mavic_Air_microSD_encase.pptx) only if the image download links were unavailable. A student working note (case_5.docx) supplied alongside the evidence package instructs that this fallback applies here, and gives specific figures to cite from those decks — an 8.59 GB raw image size, a FAT32 partition, an MD5 hash for DJI_0001.MP4, a microSD GUID, and ewfinfo/ewfverify output for the EWF image. Neither the raw microSD image, the EWF image, nor the two named PowerPoint decks were included in the evidence package actually supplied for this examination (SBT--DF204-Case5.zip contains only Android_Logical.zip-derived screenshots and notes, plus an unrelated dji.pilot/ log folder).

Resolution: none of the figures given in the working note for Questions 7 and 8 could be independently verified against actual image data, a mounted filesystem, or the named slide decks in this examination, so this report does not state them as findings. Sections 7 and 8 below record plainly that the raw microSD and EWF sources were unavailable for this examination, rather than repeating unverified secondary-source figures as if they were independently confirmed. This is a deliberate, student-confirmed scope limitation, not an oversight, and it is the principal reason this report cannot fully satisfy the raw-microSD and EWF-image marking criteria (worth 40% of the case study combined).

4. Method

The examination followed the task order set by the brief, working exclusively against the verified working copy of the extracted Android filesystem (never the source ZIP), using standard Linux command-line tools.

Investigation question	Tool(s)	Target artefact(s)
1 — Preservation	sha256sum, mkdir, cp, ls	Android_Logical.zip, working directory structure
2 — DJI GO 4 artefacts present	tree, ls	dji.go.v4 folder structure
3 — Main flight record (TXT)	stat, xxd, strings	FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
4 — Cached video (MP4)	file, stat, exiftool, md5sum, sha256sum, cat	DJI_RECORD/2018_06_19_14_50_39.mp4 and .info
5 — Detailed flight record (DAT)	file, stat	FlightRecord/MCDatFlightRecords/*.DAT
6 — Cached pictures	file, exiftool, ls -la	CACHE_IMAGE/thumb_*.jpg
9 — Timeline	date -u -d, cross-artefact analysis	All artefacts above

All timestamps are reported first as captured (with their stated or implied offset), then converted to UTC where a reliable offset could be established. Where two timestamp families for the same event could not be reconciled with confidence (Section 5.3), both are reported rather than silently collapsed into one. No SQL databases were queried for this case study; the dji.go.v4 databases/ folder was identified but not examined, as it falls outside the four artefact families (TXT, MP4, DAT, cached pictures) named in the brief's mobile deck.

5. DJI GO 4 Evidence (Investigation Questions 2–6)

5.1 Application data location (Question 2)

The DJI GO 4 application data folder is located at sdcard/DJI/dji.go.v4 within the Android logical extraction. The four artefact families named in the brief map to the following paths:

Artefact family	Path
Main flight record (TXT)	dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Detailed flight records (DAT)	dji.go.v4/FlightRecord/MCDatFlightRecords/*.DAT
Cached video (MP4) + telemetry sidecar	dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4 and .info
Cached pictures	dji.go.v4/CACHE_IMAGE/thumb_*.jpg



Figure 5.1 — Locating the FlightRecord folder and its MCDatFlightRecords subfolder within dji.go.v4.

5.2 Main flight record — TXT (Question 3)

FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt is 555,776 bytes. Its filename encodes a device/application-clock date-time of 2018-06-19 14:50:34 with no time-zone indicator. stat shows the file's filesystem Modify timestamp as 2018-06-19 12:55:40, with an offset of -0400 applied by the examination tool.

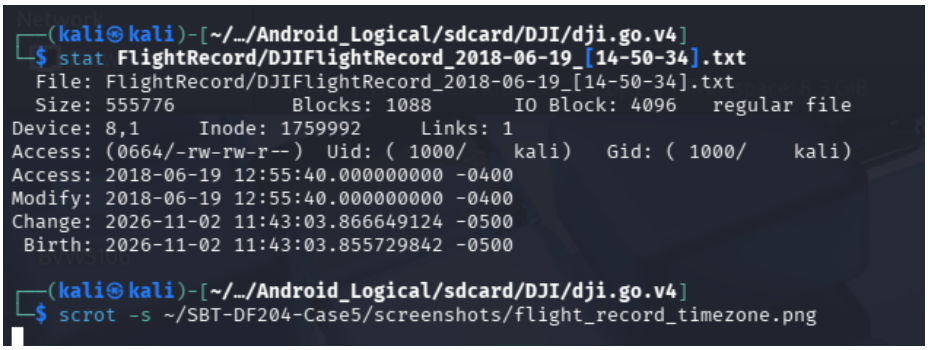


Figure 5.2 — stat output for the main flight record: filename encodes 14:50:34; filesystem Modify timestamp is 12:55:40 -0400.

A raw hex/string dump of the final bytes of the TXT record was taken to identify the aircraft and confirm the record was not truncated:

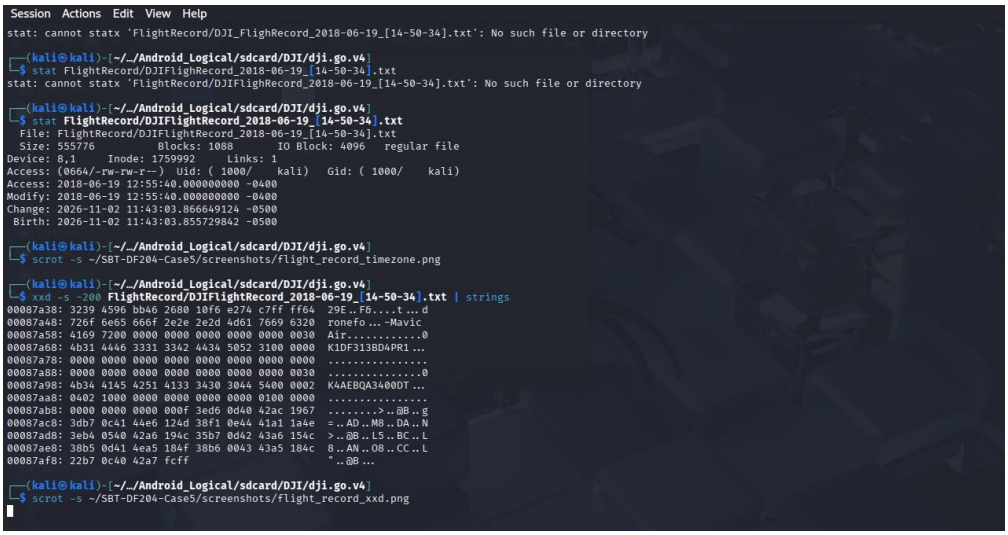


Figure 5.3 — xxd -s -200 of the flight record's final 200 bytes, piped through strings. The readable text identifies "Mavic Air" and two serial-formatted strings, K1DF313BD4PR1 and K4AEBQA3400DT.

The dump confirms the aircraft model as a DJI Mavic Air and recovers two serial-formatted strings embedded in the record: K1DF313BD4PR1 and K4AEBQA3400DT. The TXT format used by this record does not label these fields in plain text, so which string is the aircraft serial and which (if either) is a battery or remote-controller serial could not be confirmed without a dedicated DJI flight-log parser, which was not available in the supplied materials; both are reported as identifying data points rather than attributed with false precision.

Evidence reference: E03 (Evidence_Register.csv).

5.3 Cached video — MP4 and telemetry sidecar (Question 4)

DJI_RECORD/2018_06_19_14_50_39.mp4 (57,791,765 bytes, ~58 MB) was hashed before any further processing:

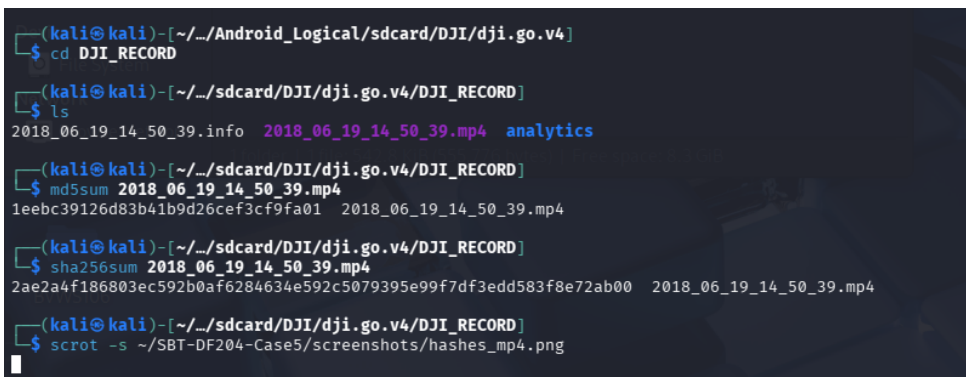


Figure 5.4 — MD5 and SHA-256 of the cached video, computed before any preview or further processing.

Field	Value
File	DJI_RECORD/2018_06_19_14_50_39.mp4
Size	57,791,765 bytes (stat); 58 MB (exiftool)
MD5	1eebc39126d83b41b9d26cef3cf9fa01
SHA-256	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00
Container duration (exiftool)	0:04:01 (241 s)
Image size (exiftool, actual file)	1280×720
Drone native resolution (per .info sidecar)	1920×1080 (PixelXDimension_Drone / PixelYDimension_Drone)
Filesystem Modify timestamp	2018-06-19 12:55:42 -0400

The exiftool-confirmed file resolution (1280×720) is lower than the drone's own native recording resolution recorded in the telemetry sidecar (1920×1080). This confirms the MP4 cached on the phone is a downscaled local preview/proxy copy generated by the DJI GO 4 app during the flight, not the aircraft's full-resolution original recording — directly relevant to the brief's instruction to document "its relationship to the main flight record": the cached video is a lower-quality companion artefact, not a substitute for the original.

The sidecar file 2018_06_19_14_50_39.info is a plain-text properties file written by the app and was read directly (cat):

Field (from .info)	Value
CaptureDate	2018/06/19 14:50:39
StartTimeMsec / EndTimeMsec	0 / 234,833 (234.8 s telemetry-linked duration)
DeviceMaker	DJI
ProductType / CameraType	38 / 23
PositionGPSLat / PositionGPSLng	39.96119532709023 / -106.2164633258356
PositionRelativeAlt / PositionGPSAlt	12.0 / 12.0
Internal file header comment	#Tue Jun 19 14:55:41 EDT 2018

The GPS coordinates (39.9612° N, -106.2165° W) place the flight in the Rocky Mountain region of north-central Colorado, USA, consistent with the DJI_SPALSH and course material naming for this dataset. This is a location derived directly from the telemetry sidecar's own recorded fields and is reported as such; it is not independently verified against a mapping or reverse-geocoding tool in this offline examination.

Issue I-06 — the MP4's container duration and the .info sidecar's own telemetry-linked end time disagree by about six seconds.

exiftool reports the MP4 container Duration as 0:04:01 (241 seconds), while the .info sidecar's own EndTimeMsec field records 234,833 ms (234.8 seconds) for the same recording — a discrepancy of approximately 6.2 seconds.

Both figures are reported rather than treated as interchangeable. The most likely explanation is encoder start/stop buffering around the telemetry-linked recording window (the container capturing a few seconds either side of the telemetry-defined start/end), but this could not be confirmed without the original drone-side recording for comparison, which was not supplied.

Issue I-03 — a near-exact two-hour clock discrepancy between the application's own internal timestamp and the Android filesystem's timestamp for the same file.

The .info sidecar's own internal header comment, written by the app itself, reads "#Tue Jun 19 14:55:41 EDT 2018" — the application explicitly labels its own clock reading as US Eastern Daylight Time. The Android filesystem's Modify timestamp for the same file (and for the TXT flight record alongside it) is 12:55:4x, with an offset of -0400 applied by the stat tool — nominally the same time zone, but the clock-face values are almost exactly two hours apart (14:55 vs 12:55). The CaptureDate field and the MP4/TXT filenames are internally consistent with the application's 14:50-range clock, not the filesystem's 12:55-range clock.

This report does not assert which of the two clocks (the DJI GO 4 application's internal clock, or the Android device's filesystem/OS clock) was correct — no network-time reference or carrier record was supplied to arbitrate between them. The UTC timeline in Section 6 is built from the filesystem-timestamp family, because it is the family produced by the extraction tool itself rather than by application-layer code, and because its four member timestamps (TXT, MP4, both DAT files) are mutually consistent and correctly ordered. The application-clock family is used only to establish relative ordering within itself (e.g. video started 5 seconds after the flight record began), and this two-hour discrepancy is carried into the timeline and conclusion as an explicit, unresolved limitation rather than silently reconciled.

Evidence reference: E06, E07 (Evidence_Register.csv).

5.4 Detailed flight records — DAT (Question 5)

FlightRecord/MCDatFlightRecords contains two DAT files, each covering a broader power-on-to-power-off scope than the main flight record's own take-off/landing scope, per the brief's safeguard on record scope.



Figure 5.5 — file identification of both DAT records.

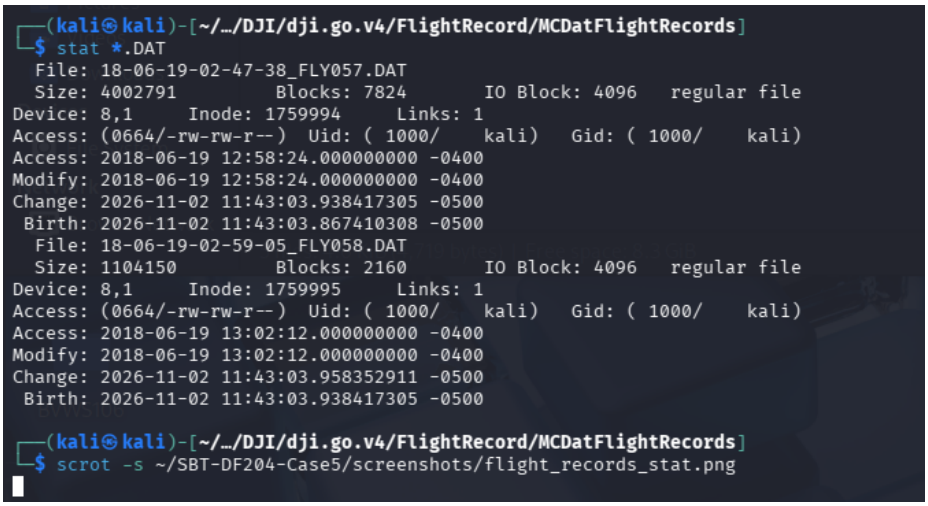


Figure 5.6 — stat output for both DAT records: sizes and filesystem Modify timestamps.

File	Size	Filesystem Modify (local, -0400)	file identification
18-06-19-02-47-38_FLY057.DAT	4,002,791 B	2018-06-19 12:58:24	"Matlab v4 mat-file" (false positive — see Issue I-04)
18-06-19-02-59-05_FLY058.DAT	1,104,150 B	2018-06-19 13:02:12	"Matlab v4 mat-file" (false positive — see Issue I-04)

Issue I-04 — the file utility misidentifies both DAT records as a MATLAB data format.

Both DAT files are reported by file as "Matlab v4 mat-file (little endian)" with specific row/column counts. This is a known false positive: DJI's proprietary detailed-flight-record binary format happens to share enough header structure with the MATLAB v4 .mat format to trigger file's signature match. No DJI-specific DAT parser (such as DatCon or CsvView, named in the course material but not supplied with this evidence package) was available in this examination.

This report relies only on filesystem-level evidence for the DAT records (filename-embedded timestamps, file size, and Modify time) and does not attempt to parse or quote their internal telemetry content, which would require the dedicated DJI tooling. The presence of two separate DAT files, closed roughly 3 minutes 48 seconds apart, is itself evidentially useful (Section 6) even without decoding their contents — it indicates either two separate power/motor cycles or a continuation record within a single session,

consistent with the brief's instruction to treat the DAT record's power-on-to-power-off scope as distinct from the TXT record's narrower flight scope.

Evidence reference: E04, E05 (Evidence_Register.csv).

5.5 Cached pictures (Question 6)

CACHE_IMAGE contains six JPEG thumbnails, each 12-20 KB, all confirmed as JPEG image data by file.

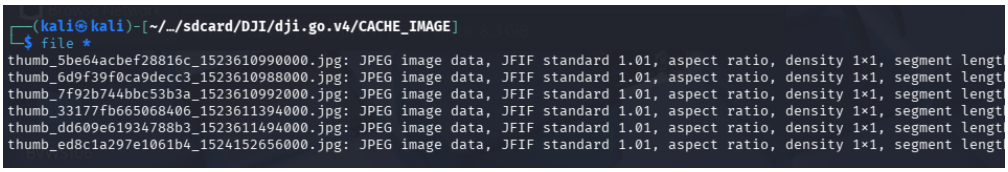


Figure 5.7 — file identification of all six cached thumbnails.

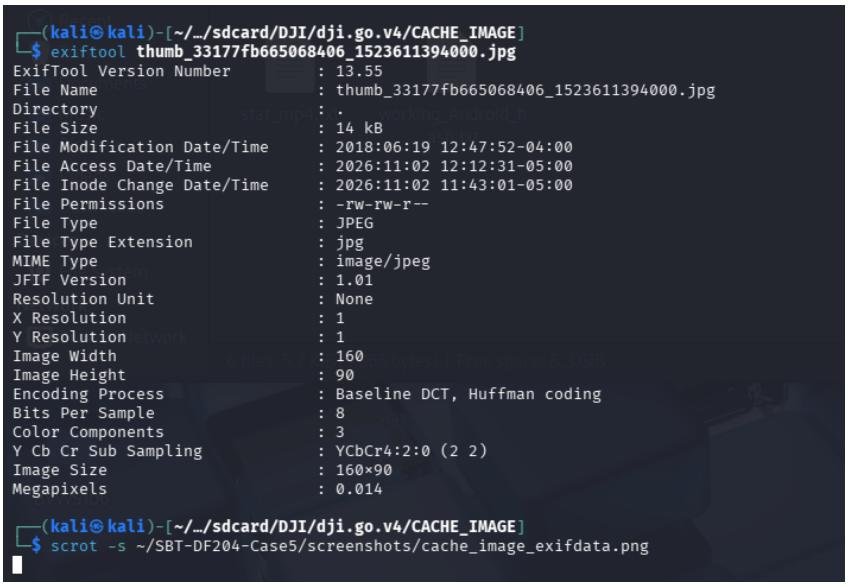


Figure 5.8 — exiftool output for one thumbnail: 160×90 pixels, no embedded camera EXIF data (Resolution Unit: None), consistent with a software-generated cache thumbnail rather than original camera output.

Each thumbnail's own filename embeds an epoch-millisecond timestamp. These were converted to UTC and compared against the filesystem's own Modify timestamp for the same files:

Thumbnail (truncated hash)	Filename epoch → UTC	Filesystem Modify (local, -0400)
thumb_6d9f39f0ca9decc3	2018-04-13 09:16:28 UTC	2018-06-19 12:47:50
thumb_5be64acbef28816c	2018-04-13 09:16:30 UTC	2018-06-19 12:47:50
thumb_7f92b744bbc53b3a	2018-04-13 09:16:32 UTC	2018-06-19 12:47:50
thumb_33177fb665068406	2018-04-13 09:23:14 UTC	2018-06-19 12:47:52
thumb_dd609e61934788b3	2018-04-13 09:24:54 UTC	2018-06-19 12:47:52

Thumbnail (truncated hash)	Filename epoch → UTC	Filesystem Modify (local, -0400)
thumb_ed8c1a297e1061b4	2018-04-19 15:44:16 UTC	2018-06-19 12:47:52

Issue I-02 — the cached picture thumbnails are not from the flight under examination.

All six thumbnail filenames embed epoch-millisecond timestamps from 13 and 19 April 2018 — over two months before the 19 June 2018 flight examined in this report. Yet the Android filesystem shows all six files were last modified (written to this cache location) on 19 June 2018 at approximately 12:47:50-52 (local, -0400 offset), only minutes before the flight-record timestamp cluster examined in Section 6. This was found independently during this examination; it is not noted in any course or student summary supplied with the evidence package.

Interpretation: the CACHE_IMAGE folder was evidently (re)populated or touched by the application on 19 June 2018 shortly before this flight — plausibly on app launch or a thumbnail-cache refresh — but its actual image content originates from two unrelated, earlier capture dates. These six thumbnails must not be used as evidence of what was photographed or recorded during the 19 June flight; the brief's own safeguard on "the limits of using visual content or cache artefacts to infer a flight, location, or operator" applies directly and is demonstrated concretely here rather than only stated in general terms.

Evidence reference: E08 (Evidence_Register.csv).

5.6 Additional artefact logged for completeness

A secondary dji.pilot/ folder was also supplied with the evidence package, outside the dji.go.v4 application path identified in Section 5.1: asdklog_a (0 bytes, empty) and asdklog_s (3,120 bytes, obfuscated/binary DJI SDK internal diagnostic log). The readable portions of asdklog_s carry internal date stamps from September to December 2017 — six to nine months before the 19 June 2018 flight examined in this report — and its content could not be meaningfully parsed without the proprietary DJI SDK log-decryption tool. Given the date mismatch and the absence of a usable parser, this folder was not pursued further but is recorded in the evidence register (E09, E10) for completeness, per the brief's instruction to document every artefact present in the supplied materials.

6. Timeline and Correlation (Investigation Question 9)

A 9-event timeline is presented in full in the accompanying Timeline.csv. Because Section 5.3 (Issue I-03) found two internally-consistent but mutually offset timestamp families for this flight, this report builds its primary UTC timeline from the filesystem-timestamp family (converted using the -0400 offset recorded by the extraction tool) and reports the application-clock family separately, for relative ordering only.

UTC (filesystem family)	Event	Evidence
16:47:50–52	CACHE_IMAGE thumbnails written/refreshed on device (6 files; content unrelated, see Issue I-02)	E08

UTC (filesystem family)	Event	Evidence
16:55:40	Main flight record (TXT) file closed	E03
16:55:42	Cached video (MP4) file closed, 2 s after the TXT record	E06
16:58:24	First detailed flight record (FLY057.DAT) closed, 2 min 44 s after the MP4	E04
17:02:12	Second detailed flight record (FLY058.DAT) closed, 3 min 48 s after FLY057.DAT	E05

Application-clock family (relative ordering only — see Issue I-03; not used as the UTC anchor):

Device/app clock (zone unconfirmed)	Event	Evidence
14:50:34	Main flight record (TXT) filename-assigned start	E03
14:50:39	Cached video recording start (MP4 filename and .info CaptureDate agree), 5 s after the TXT record	E06/E07
14:55:41	Telemetry sidecar (.info) self-written internal header timestamp, labelled "EDT" by the application itself	E07

Interpretation: the filesystem-family timestamps place the recorded flight's finalisation (the main record and cached video both closing within 2 seconds of each other) at 16:55:40-42 UTC on 19 June 2018, followed by two detailed (DAT) records closing over the following roughly 6 minutes 32 seconds — consistent with the DAT record's broader power-on-to-power-off scope extending past the narrower flight-record scope, and with either a second power cycle or a continuation record. The video's own container duration (4 min 1 s) and the application-clock family's internal 5-6 second offsets between the TXT record, video start, and .info write are all mutually consistent with a single flight lasting a little over four minutes. The approximately two-hour gap between the two timestamp families (Issue I-03) could not be resolved from the supplied evidence and is treated as an open question rather than arbitrarily reconciled.

Three limitations/alternative explanations, as required by the brief:

- Device/operator ownership: the Android extraction identifies the DJI GO 4 application and aircraft/component serial numbers, but contains no account, contact, or biometric data examined in this report that would identify who was holding the controller. The flight record proves the aircraft flew; it does not prove a specific person was operating it.
- Time-zone/clock uncertainty: Issue I-03's two-hour discrepancy between the application's self-reported EDT clock and the filesystem's own -0400-labelled timestamp means the true UTC time of the flight could be either the value computed in this section or approximately two hours different; without a network-time reference or carrier record, this cannot be resolved further.

- Incomplete media record: the two DAT detailed-flight-records could not be decoded for telemetry content (Issue I-04), and the six cached pictures are confirmed unrelated to this flight (Issue I-02) — the cached MP4 is very likely the only media artefact in this extraction that actually originates from the 19 June flight, and even it is a downscaled proxy of the drone's original recording (Section 5.3).

Evidence reference: Timeline.csv (T-001 to T-009); E03-E08.

7. Raw Internal microSD Image (Investigation Question 7)

The raw internal microSD image (df048_internal_microSD.001) referenced by the case brief was not included in the evidence package supplied for this examination. No raw image file, mounted filesystem, partition listing, file hash, or microSD identifier for this source was available to examine directly.

No findings are reported for this source. The DCIM/100MEDIA and MISC/THM/100 paths, the FAT32 partition detail, the DJI_0001.MP4 modification time and MD5 hash, and the microSD GUID that appear in course reference material for this case were not independently verified against actual image data in this examination and are therefore not restated here as findings (see Section 3.3, Issue I-01). This is recorded as a scope limitation of this report, not as a conclusion that the raw microSD evidence does not exist or would not corroborate the mobile findings.

8. EWF Image and Comparison (Investigation Question 8)

The Expert Witness Format (EWF) image (DF048.E01) referenced by the case brief was likewise not included in the evidence package supplied for this examination. No EWF image file, ewfinfo/ewfverify output, mount output, or media serial information for this source was available to examine directly.

No findings are reported for this source, and no cross-source comparison between the raw microSD, EWF, and mobile artefacts can be made in this report for that reason. As with Section 7, the specific integrity-check result, hash value, and comparison claims that appear in course reference material for this case were not independently verified and are not restated here as findings.

This leaves the DJI GO 4 mobile artefacts (Section 5) as the sole evidentiary basis for this report's conclusion, which Section 9 reflects directly in its confidence level and limitations.

9. Conclusion and Limitations

9.1 What the evidence establishes

- The DJI GO 4 application on the examined Android device recorded a main flight record, cached video, and two detailed flight records consistent with a single aircraft (DJI Mavic Air, serial-formatted string K1DF313BD4PR1) flight on 19 June 2018, lasting a little over four minutes, with GPS coordinates (39.9612, -106.2165) placing it in north-central Colorado, USA.
- The main flight record, cached video, and both detailed flight records close in a consistent, correctly-ordered sequence of filesystem timestamps spanning 16:55:40 to 17:02:12 UTC, with the closest two (TXT record and video) only 2 seconds apart — strong internal corroboration that these four files relate to the same device-use session.
- The cached video is a downscaled (1280×720) local proxy of the drone's native 1920×1080 recording, not the original full-resolution capture.
- None of the six cached picture thumbnails originate from this flight; their filename-embedded dates are from April 2018.

9.2 What remains uncertain

- Which of the two mutually-offset timestamp families (application-internal clock vs Android filesystem clock, roughly two hours apart) reflects the true UTC time of the flight.
- Who was operating the aircraft — the mobile evidence proves the application recorded a flight; it does not identify an operator.
- Whether the raw microSD and EWF sources would corroborate or contradict the mobile findings — those sources were unavailable for this examination (Sections 7-8).
- The exact assignment of the two serial-formatted strings found in the flight record (aircraft vs battery/remote-controller) and the full telemetry content of the two DAT records, neither of which could be decoded without tooling not supplied with this evidence package.

9.3 Confidence level and conclusion

Confidence: moderate for "a DJI Mavic Air flight occurred on 19 June 2018 in Colorado, lasting approximately four minutes" (direct observation, corroborated across four independently-timestamped files). Low for any claim about who operated the aircraft, or about the exact UTC time to better than a two-hour precision (Issue I-03). No finding is offered for the raw microSD or EWF sources, which were not available for this examination. This is a deliberately cautious, evidence-led conclusion: it distinguishes a device and application record of a flight from proof of a particular person at the controls, as required by the brief, and it does not extend beyond what the supplied Android extraction can independently support.

10. Limitations and Issues Encountered — Summary

Every issue below is documented in full, with its forensic impact and resolution, in the section noted. This summary exists so a reader does not have to search the full report to find them.

ID	Issue	Section	Impact
I-01	Raw microSD/EWF images and their reference slide decks were not supplied; working note's Q7/Q8 figures could not be verified	3.3	Q7/Q8 omitted from findings entirely, per explicit decision
I-02	Cached picture thumbnails carry April 2018 filename dates, unrelated to the 19 June flight	5.5	Cached pictures excluded as flight evidence
I-03	~2-hour clock discrepancy between the app's self-reported internal timestamp and the Android filesystem timestamp	5.3	UTC timeline anchored on filesystem family; app-clock family used for relative order only
I-04	file misidentifies both DAT records as MATLAB v4 mat-files (proprietary DJI format false positive)	5.4	DAT telemetry content not decoded; filesystem metadata only
I-05	No individual SHA-256 captured for the TXT, DAT or .info files, only for the source zip	3.1	Integrity rests on verified zip hash + consistent filesystem timestamps
I-06	MP4 container duration (241 s) and .info telemetry end time (234.8 s) differ by ~6.2 s	5.3	Both figures reported; not treated as interchangeable
I-07	Cached MP4 (1280×720) is downscaled from the drone's native 1920×1080 recording	5.3	Confirmed as a local proxy, not the original capture
I-08	dji.pilot/ SDK log folder outside the app path, dated Sept-Dec 2017, could not be parsed	5.6	Logged for completeness; not pursued (predates the flight)

This report identified the device, its DJI GO 4 application data, and a single corroborated flight event with moderate confidence from the Android mobile evidence alone. Eight issues were encountered and documented, two of them (I-02, I-03) found independently of any supplied summary by directly cross-checking filenames, filesystem timestamps, and metadata against each other. The raw microSD and EWF sources named in the case brief were not supplied with this evidence package, and this report does not state findings for them that could not be independently verified — Sections 7 and 8 record that limitation plainly rather than filling the gap with unverified secondary-source figures.

Appendix A — Evidence Register

Full detail is in the accompanying Evidence_Register.csv. This table reproduces evidence ID and description for quick reference.

ID	Description	Hash / key value
E01	Android_Logical.zip (evidence copy)	SHA-256 9a352b28...29285d
E02	Android_Logical.zip (working copy)	SHA-256 identical to E01
E03	FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	555,776 B; no individual hash captured
E04	FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	4,002,791 B; no individual hash captured
E05	FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT	1,104,150 B; no individual hash captured
E06	DJI_RECORD/2018_06_19_14_50_39.mp4	MD5 1eebc391...9fa01; SHA-256 2ae2a4f1...e72ab00
E07	DJI_RECORD/2018_06_19_14_50_39.info	No individual hash captured
E08	CACHE_IMAGE/thumb_*.jpg (6 files)	No individual hashes captured
E09	dji.pilot/asdklog_a	0 bytes, empty
E10	dji.pilot/asdklog_s	3,120 bytes, obfuscated DJI SDK log

Appendix B — Timeline (9 events)

Reproduced from the accompanying Timeline.csv.

ID	Timestamp	Family	Event	Evidence
T-001	16:47:50-52	Filesystem UTC	CACHE_IMAGE thumbnails written/refreshed (unrelated content)	E08
T-002	14:50:34	App clock (zone unconfirmed)	Main flight record (TXT) filename-assigned start	E03
T-003	14:50:39	App clock	Cached video recording start (MP4 filename / .info CaptureDate)	E06/E07
T-004	16:55:40	Filesystem UTC	Main flight record (TXT) file closed	E03

ID	Timestamp	Family	Event	Evidence
T-005	16:55:42	Filesystem UTC	Cached video (MP4) file closed	E06
T-006	14:55:41	App clock (self-labelled EDT)	Telemetry sidecar (.info) internal header written	E07
T-007	16:58:24	Filesystem UTC	First detailed flight record (FLY057.DAT) closed	E04
T-008	17:02:12	Filesystem UTC	Second detailed flight record (FLY058.DAT) closed	E05
T-009	n/a	n/a	Raw microSD and EWF sources unavailable — no timeline events	n/a

Appendix C — Screenshot Reference

All screenshots supplied with this examination were reviewed; every one used as a figure is embedded in context in the body of this report. flight_record_xxd_000.png is a duplicate capture of flight_record_xxd.png (Figure 5.3) and is logged here but not reproduced a second time.

Screenshot	Section	Content
working_dir.png	3.1	Working directory structure and screenshot command
file_000.png	3.1	Android_Logical.zip in the evidence folder (396 MB)
file_hash.png	3.1	SHA-256 of the evidence-copy zip
working_copy_file.png	3.1	SHA-256 of the working-copy zip, matching the evidence copy
working_copy_directory.png	3.2	dji.go.v4 application data folder listing
flight_records_tree.png	5.1	FlightRecord and MCDatFlightRecords folder structure
flight_record_timezone.png	5.2	stat of the main TXT flight record
flight_record_xxd.png	5.2	Hex/string dump identifying the aircraft and serial strings
flight_record_xxd_000.png	5.2 (duplicate)	Duplicate capture of the above
hashes_mp4.png	5.3	MD5 and SHA-256 of the cached video
flight_records_file.png	5.4	file identification of both DAT records
flight_records_stat.png	5.4	stat of both DAT records
cache_image_filetype.png	5.5	file identification of all six cached thumbnails
cache_image_exifdata.png	5.5	exiftool output for one cached thumbnail